

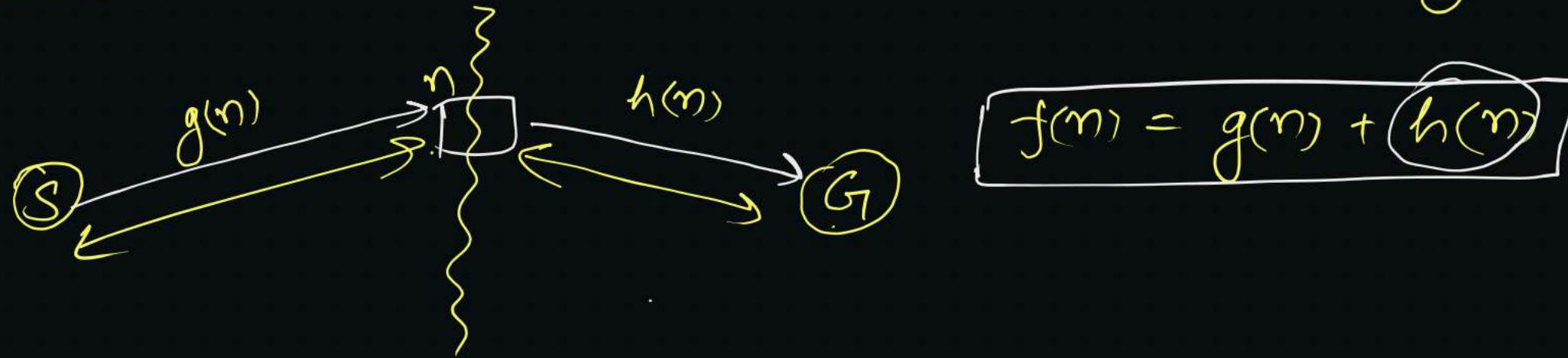
Performance analysis :-

↳ Admissibility :-

↳ An admissible heuristic is one that never overestimates the cost to reach the goal state.

$$\forall n, \underline{h(n)} \leq \underline{h^*(n)}$$

where, $h^*(n)$ = optimal cost from node n to the goal state.



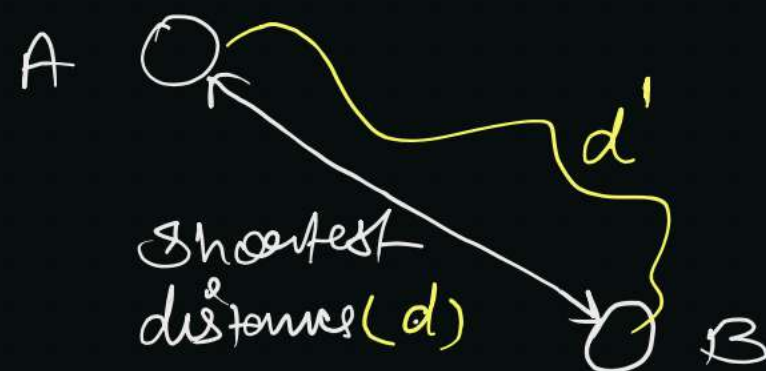
↳ Thus, an admissible heuristic is optimistic.

↳ With an admissible heuristic, A^* is optimal. (yes)

↳ If $\exists n, h(n) > h^*(n)$, then $h(n)$ is not admissible.

↳ eg. straight line distance heuristic is admissible.

is L_1 norm admissible?



$$\boxed{d' \geq d}$$

we can't build a road having distance less than d .

proof :-

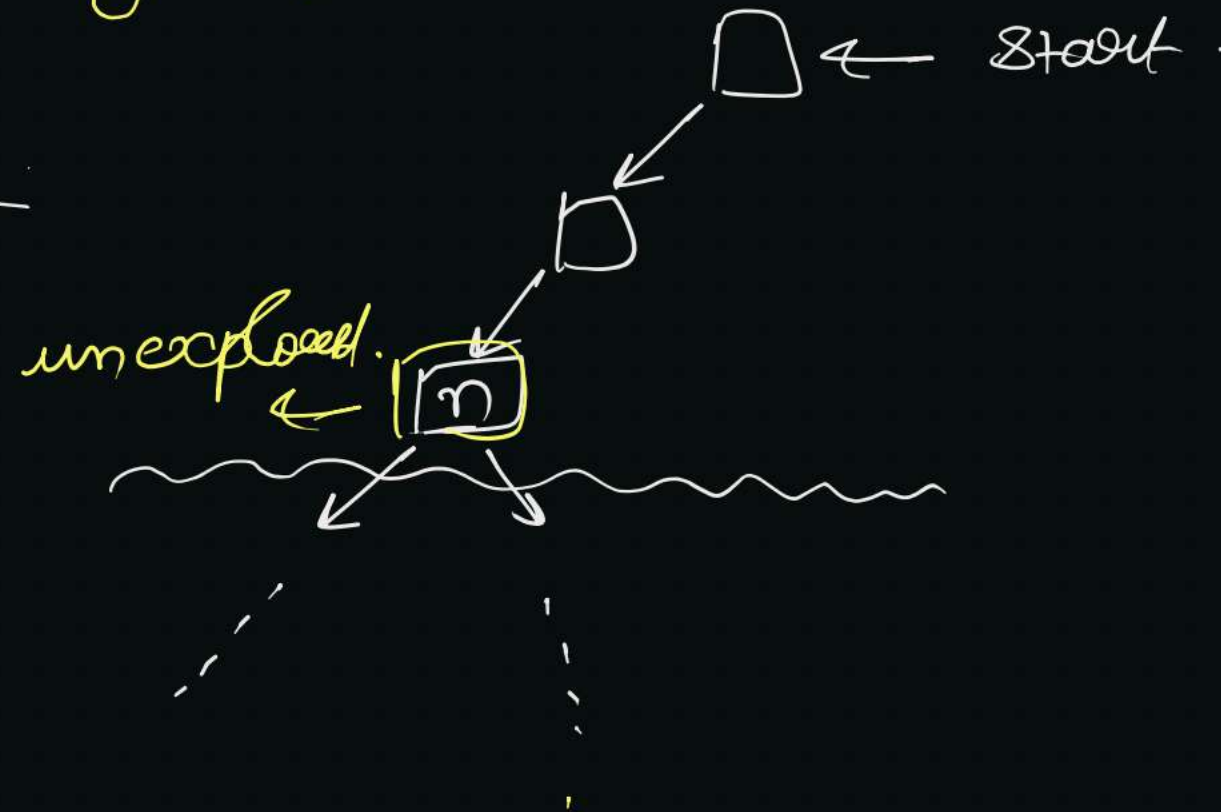
Suppose we have an optimal path with cost c^* .
but the algorithm (A^*) returns a path with $\boxed{c > c^*}$.

Let there be a node " n ", in the optimal path,
which remain unexplored.
else we would have got optimal cost (c^*).

$$\Rightarrow \boxed{f(n) > c^*} \text{ --- (1)}$$

priority			
	...	(goal, c^*)	...
			(n, $f(n)$)
			...

If (n, $f(n)$) is present after
(goal, c^*) $\rightarrow$ then it



For any node n ,

□ goal.

$$\Rightarrow f(n) = \underline{g(n)} + \underline{h(n)} ; \left[\underline{g(n)} = \underline{g^*(n)} \rightarrow \text{as } n \text{ lies on the optimal path.} \right]$$

$$\Rightarrow f(n) = g^*(n) + h(n) \rightarrow$$

$$\Rightarrow f(n) \leq \underline{g^*(n) + h^*(n)} \rightarrow \left[\forall n, \underline{h(n)} \leq \underline{h^*(n)} \right] ?$$

$$\Rightarrow \underline{f(n)} \leq \underline{f^*(n)}$$

$\Rightarrow$ add $g^*(n)$ to both sides.

$$\Rightarrow h(n) + g^*(n) \leq h^*(n) + g^*(n)$$

$$\Rightarrow \boxed{f(n) \leq \underline{C^*}} \text{ --- (2)}$$

$$\Rightarrow f(n) \leq h^*(n) + g^*(n)$$

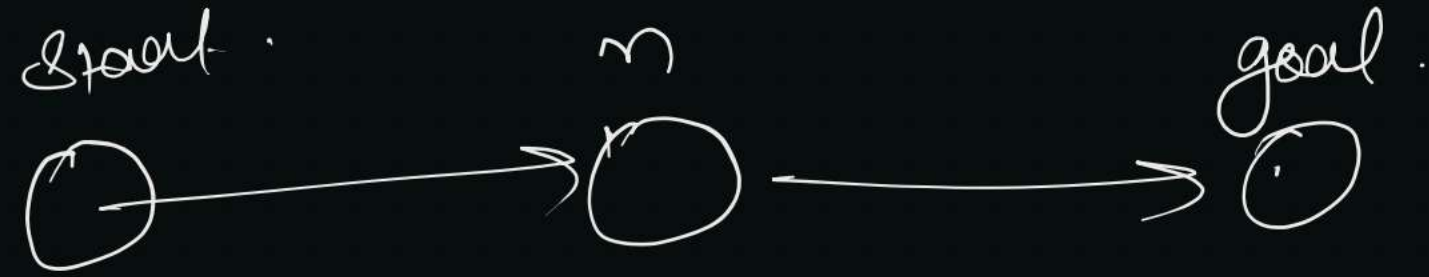
eq. (2) is contradicting eq. (1), therefore an admissible heuristic will always give an optimal path.

the supposition that the algorithm will return a suboptimal path is incorrect.

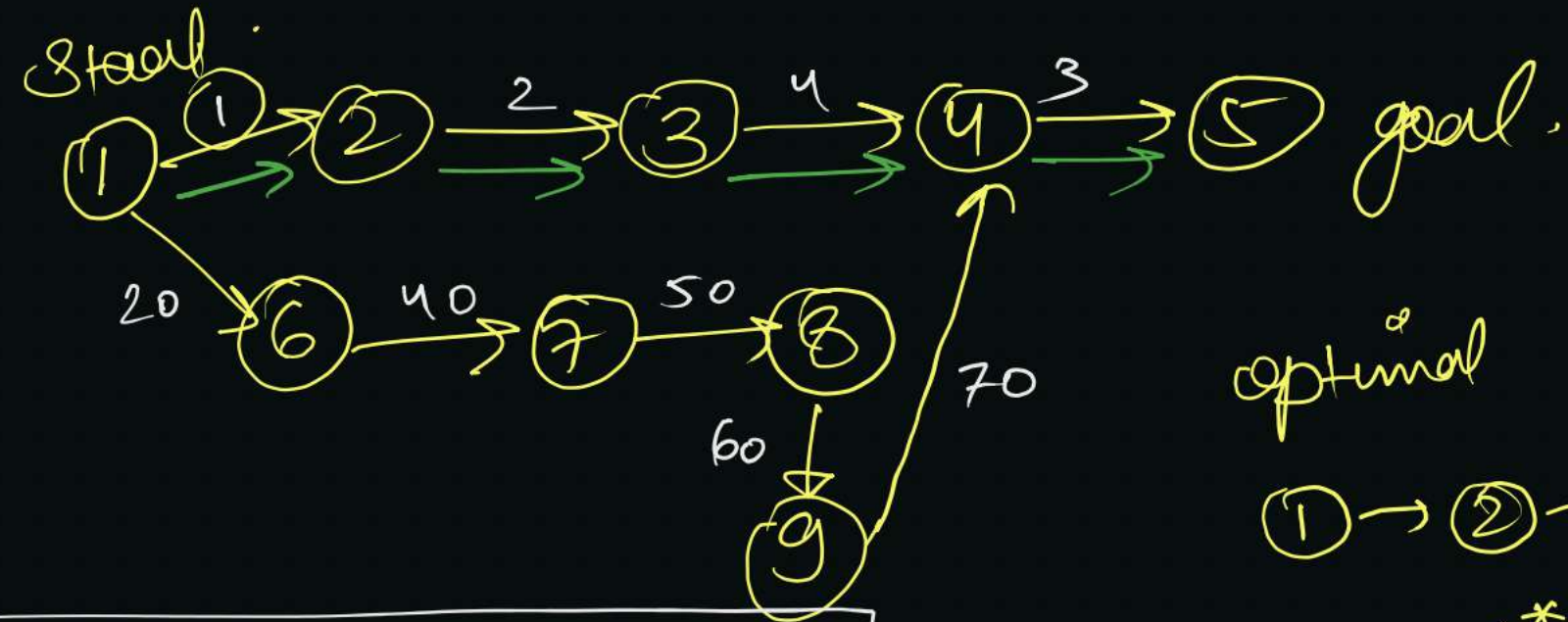
$$f(n) = g(n) + h(n) \Rightarrow f^*(n) = \underline{g^*(n)} + \overbrace{h^*(n)}^{\substack{\text{giving the remaining} \\ \text{optimal cost.}}} \quad n \rightarrow \text{goal.}$$

$\hookrightarrow$ optimal path cost from start $\rightarrow n$.

$$= C^* \text{ (optimal cost)}$$



ex.



optimal path : $C^* = 10$

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

$$n=1 \quad , \quad f^*(1) = g^*(1) + h^*(1) \\ = 0 + 10 = 10$$

$$n=2 \quad ; \quad f^*(2) = g^*(2) + h^*(2) \\ = 1 + (2+4+3) \\ = 10$$

$\vdots$

$$\forall n \in \{1, 2, 3, 4, 5\} \rightarrow$$

$$f^*(n) = g^*(n) + h^*(n) \\ = C^* \\ = 10$$

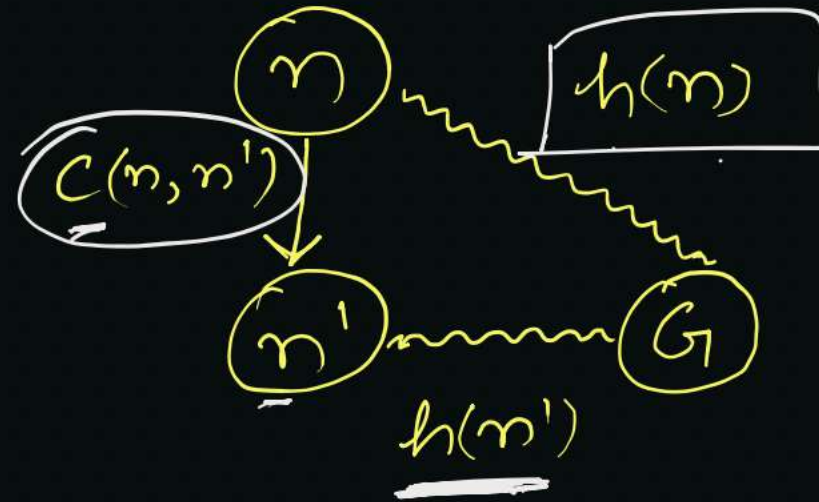
$$\# \quad \forall n, \quad f(n) \leq f^*(n)$$

$\hookrightarrow$ given $h(n)$ is admissible.

↳ Consistency :-

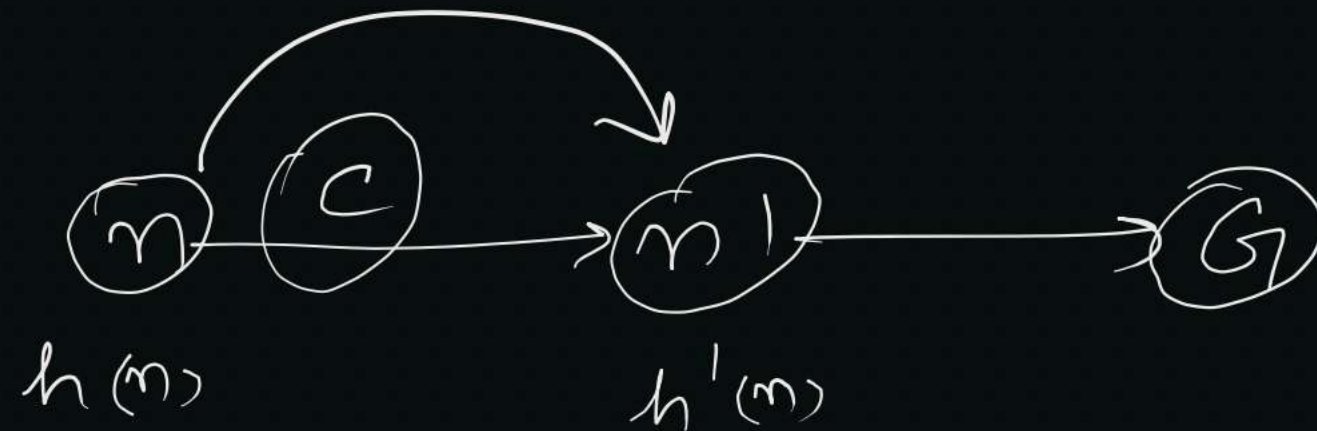
$\forall n, \forall \text{ successor } n'$

$$h(n) \leq c(n, n') + h(n')$$



↳ then, $h(n)$ is consistent.

- Every consistent heuristic is also admissible.

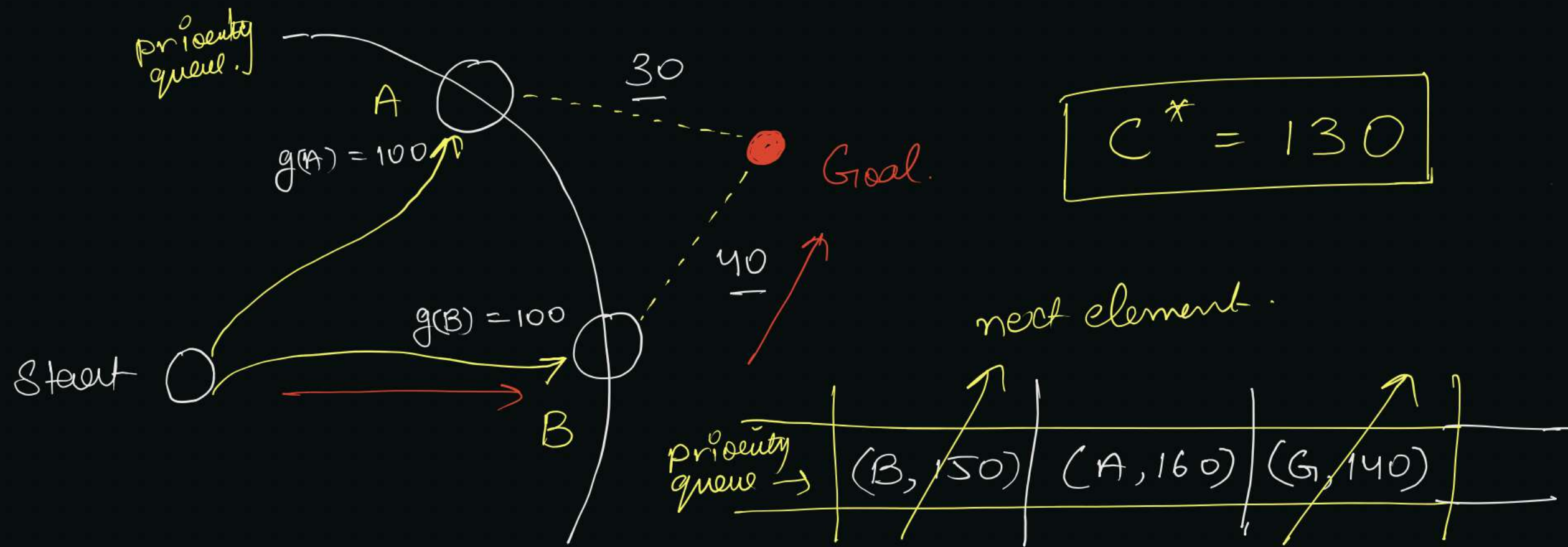


$$h(n) - h'(n) \leq c \Rightarrow h(n) \leq h'(n)$$

Q. Assuming that a perfect heuristic function, $h(n) = h^*(n)$ doesn't exist, there are only 2 possibilities left:

- ① it underestimates the distance to the goal. $\rightarrow$ optimal sol.
- ② it overestimates the distance to the goal. $\rightarrow$ suboptimal sol.

Which one is better? & why.



- assuming the heuristic to be overestimating.

$$h(A) = \underline{60} ; h(B) = \underline{50} \quad \left. \begin{array}{l} \text{the algorithm thinks} \\ B \text{ is closer than A.} \end{array} \right\}$$

$$\Rightarrow f(A) = g(A) + h(A) = 100 + 60 = 160$$

$$\& \quad f(B) = g(B) + h(B) = 100 + 50 = 150$$

- the algorithm picks B, & add it will add G₁ to the queue.

$$f(G) = g(G) + h(G)$$

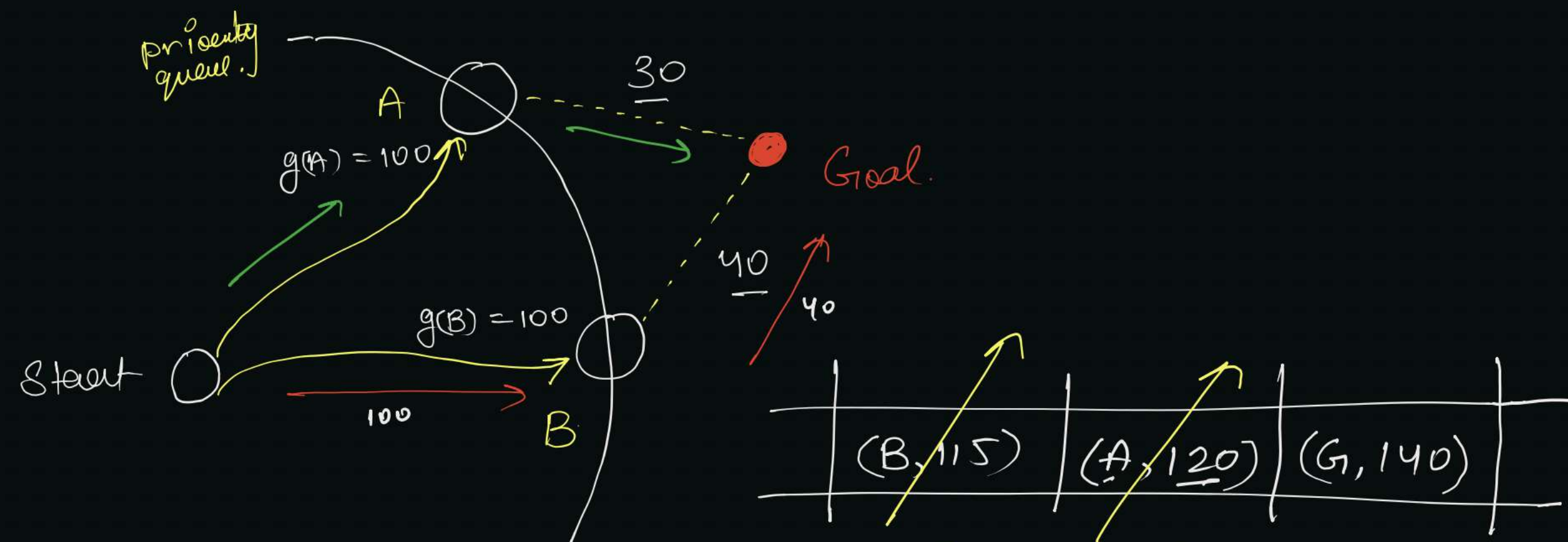
$$= (100 + 40) + 0 = 140$$

- the algorithm picks $G_1 \rightarrow \text{cost} = 140$ $\rightarrow$ Search terminates. $\left. \begin{array}{l} \text{Search} \\ \text{terminates.} \end{array} \right\}$

$\hookrightarrow$ it's a suboptimal solution.

★. Optimality is not guaranteed in case of an overestimating heuristic.

eg. In case we have underestimating heuristic.



↳ $h(A) = 20$, $h(B) = 15$ } Assuming B is more closer than A to the goal.

$$f(A) = g(A) + h(A) = 120$$

$$f(B) = g(B) + h(B) = 115$$

↳ The algorithm will pick B, & add G to the queue.

$$f(G) = g(G) + h(G) = 140 + 0 = 140$$
 } search continues.

↳ the algorithm will pick A,

$$f(G) = g(G) + h(G) = 130 + 0 = 130$$
 } Search will terminate, bcoz, we found the goal & no open node having estimated cost < 130 .

H.W.

If,

then

Consistency $\Rightarrow$ admissibility
admissibility $\Rightarrow$ consistency

?